

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

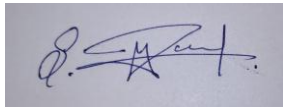
Duration :60 Minutes

Code of Conduct

I **MAGESH.S (192525002)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.

Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

Q1. Retail Data Optimization Problem

A retail chain operating both physical stores and an online platform generates large volumes of sales, inventory, and customer behavior data. It wants real-time inventory tracking, demand forecasting, and personalized marketing recommendations.

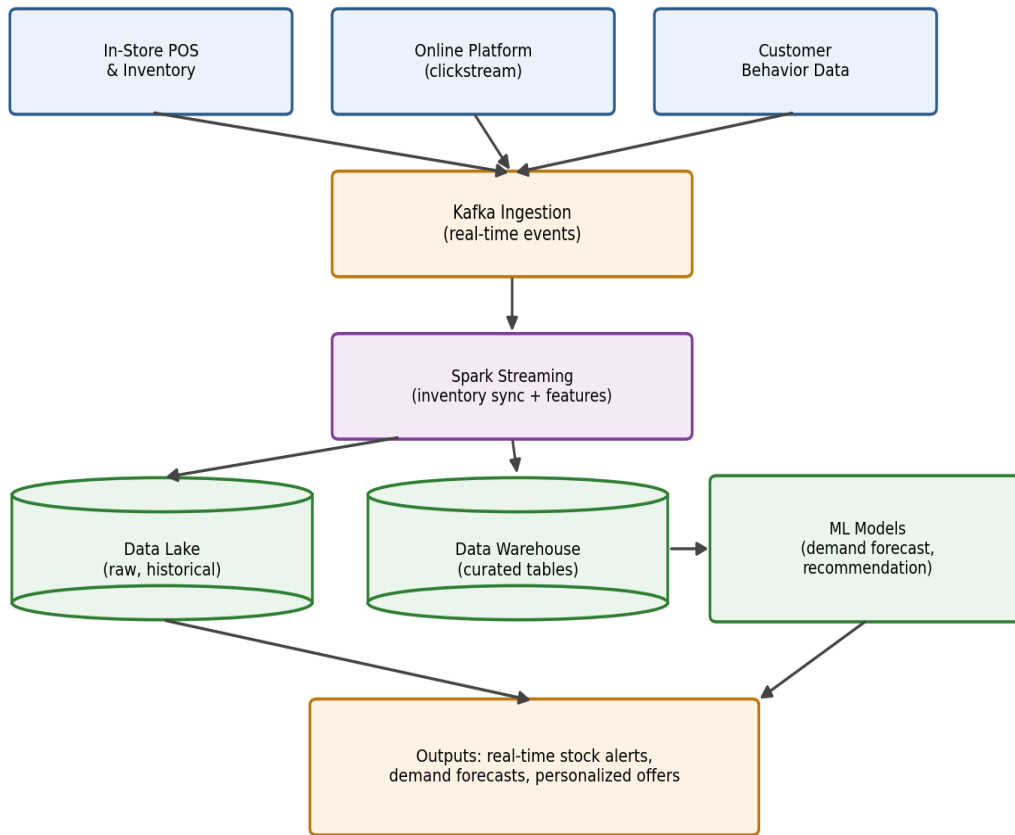
Understanding the industry context

The core challenge is unifying data from disparate sources — in-store POS systems, e-commerce platforms, and customer interaction logs — into a single pipeline that supports both operational needs (real-time stock visibility) and strategic needs (forecasting, personalization), without one channel's data lagging behind or contradicting another.

Big data architecture and tools

- Ingestion: Apache Kafka streams sales transactions, inventory updates, and clickstream events from all channels in real time.
- Stream processing: Spark Streaming (or Flink) processes incoming events to update live inventory counts and compute real-time behavioral features.
- Storage: a data lake (e.g., Amazon S3/HDFS) retains raw historical data for flexibility, while a data warehouse (e.g., Snowflake/BigQuery) holds curated, query-optimized tables for reporting and model input.
- Analytics models: time-series forecasting models (ARIMA, Prophet, or LSTM-based models) for demand forecasting; collaborative filtering or gradient-boosted models for personalized recommendations.
- Serving layer: real-time dashboards for store managers (stock alerts) and a recommendation API feeding the website/app.

Q1 – Retail Big Data Architecture



Solution design rationale

Combining stream processing (for immediate stock/alert needs) with batch/warehouse-based modeling (for deeper forecasting and personalization) balances latency requirements against the computational cost of complex models — inventory counts must update within seconds, but demand forecasts can tolerate being recomputed hourly or daily.

Security and risk awareness

Customer behavior and purchase data include personally identifiable information (PII), so access to raw customer-level data should be role-restricted, with recommendation models trained on anonymized or aggregated features where possible, and encryption applied to data at rest and in transit.

Q2. Social Media Fraud Detection Problem

A social media platform processes billions of user interactions daily (posts, comments, likes, shares) and wants to detect fake accounts and misinformation with real-time or near real-time processing.

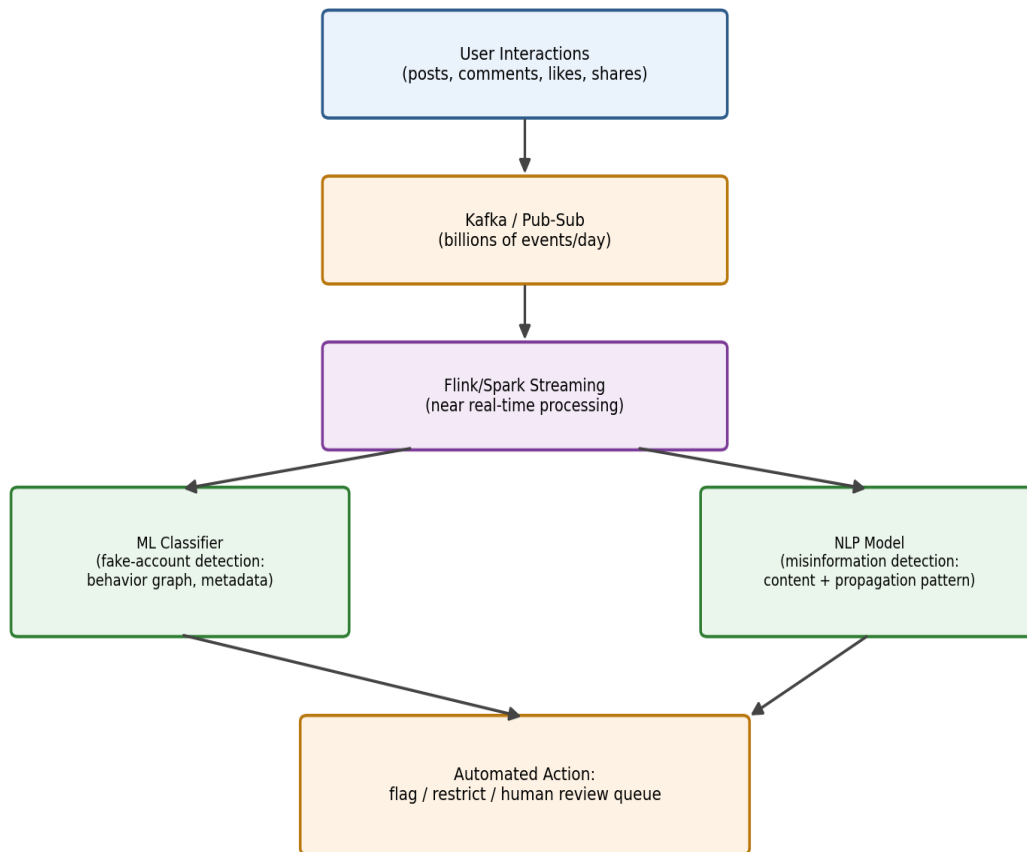
Understanding the industry context

At this scale, waiting for periodic batch analysis lets fake accounts and misinformation spread widely before detection, so the system must score and act on content and account signals within seconds to minutes of creation, while still handling the sheer volume of billions of daily events cost-effectively.

Scalable big data solution

- Ingestion: a distributed pub-sub system (Kafka or a managed equivalent) absorbs the firehose of interaction events across the platform.
- Stream processing: Apache Flink or Spark Streaming performs near real-time aggregation — computing account behavior graphs (posting frequency, follower/following ratios, interaction patterns) and content propagation metrics as events arrive.
- Machine learning — fake account detection: a classifier trained on account metadata and behavioral graph features (e.g., using graph neural networks or gradient boosting) flags accounts exhibiting bot-like or coordinated inauthentic behavior.
- Machine learning — misinformation detection: NLP models (transformer-based text classifiers) analyze content combined with propagation-pattern features (unusually fast, coordinated sharing) to flag likely misinformation.
- Action layer: flagged accounts/content are either automatically restricted or routed to a human moderation review queue, balancing automation speed with the risk of false positives on legitimate content.

Q2 — Social Media Fake-Account & Misinformation Detection



Role of ML and stream processing frameworks

Stream processing frameworks make it possible to compute the graph and propagation features that fake-account and misinformation detection actually depend on — these signals only exist "in motion" (how fast something spreads, how an account behaves over time) and are lost if data is only analyzed in daily batches. Machine learning is essential because manually written rules cannot keep pace with rapidly evolving evasion tactics used by bad actors; models can be retrained continuously as new patterns emerge.

Security and risk awareness

False positives (wrongly flagging genuine users) carry real reputational and free-speech risk, so a human-in-the-loop review step is important for borderline cases; the system should also log all automated actions for auditability and be resistant to adversarial manipulation (e.g., attackers probing the model to learn how to evade it).

Q3. Government Data Platform Problem

A government agency is building a national data platform combining census, economic, and geospatial datasets, with multiple departments needing shared, integrated access, strict interoperability requirements, and strict privacy and security requirements.

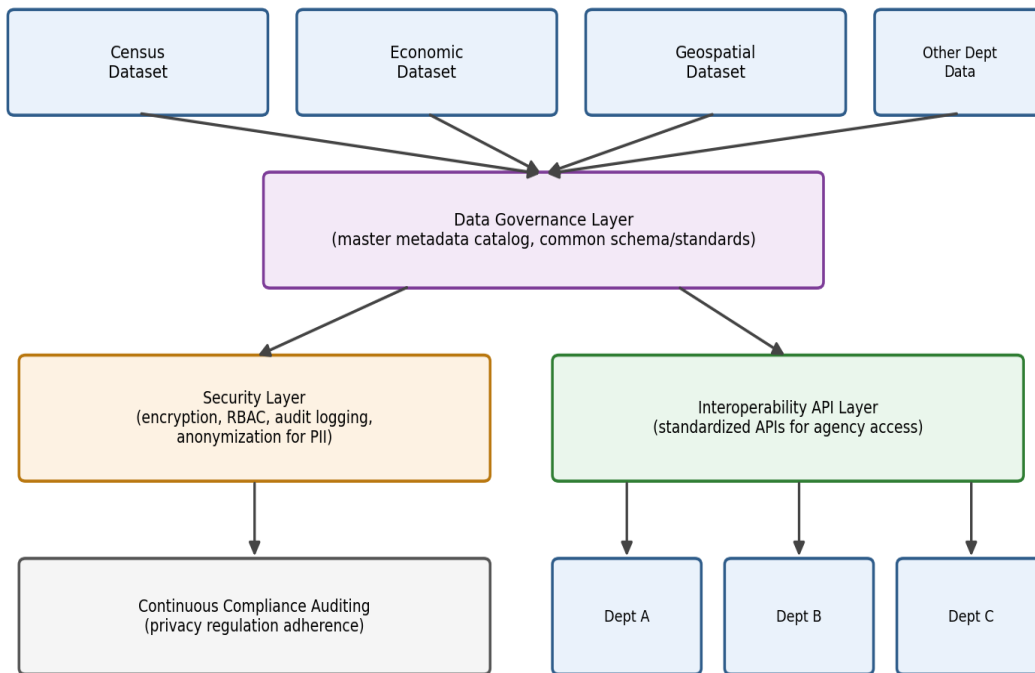
Understanding the industry context

Government data platforms face a distinct tension: different agencies each need broad analytical access to combined datasets for policy-making, yet the underlying data (especially census and economic records) is highly sensitive, and legal frameworks strictly limit who can see identifiable information — the governance strategy must satisfy both needs simultaneously.

Governance and data management strategy

- Unified data governance layer: a master metadata catalog and common data standards/schema ensure census, economic, and geospatial datasets can be meaningfully joined and interpreted consistently across departments.
- Interoperability API layer: standardized APIs (rather than direct database access) expose data to departments, decoupling each agency's internal systems from the platform's underlying storage and enabling controlled, auditable access.
- Security layer: field-level encryption, role-based access control (RBAC) scoped to each department's mandate, and anonymization/aggregation of personally identifiable census data before it's exposed to departments that don't need identifiable records.
- Data classification policy: explicitly categorize datasets (public, restricted, confidential) so access rules can be applied systematically rather than ad hoc.
- Continuous compliance auditing: ongoing monitoring and logging of data access against privacy regulations, with periodic audits to verify departments are not exceeding their authorized scope.

Q3 — National Data Platform Governance Architecture



Solution design rationale

Routing all cross-department access through a governed API layer (instead of direct data-store access) means privacy rules, anonymization, and auditing are enforced centrally and consistently, rather than depending on every department correctly implementing its own controls.

Security and risk awareness

Re-identification risk is a major concern when combining census, economic, and geospatial data, since seemingly anonymized records can sometimes be re-identified when joined with other datasets — the governance strategy must include techniques like k-anonymity or differential privacy for any cross-dataset analytical access, plus strict logging to detect misuse.

Q4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute and needs to detect fraudulent activity in real time, analyzing both historical and streaming transaction data, with low latency and high accuracy as critical requirements.

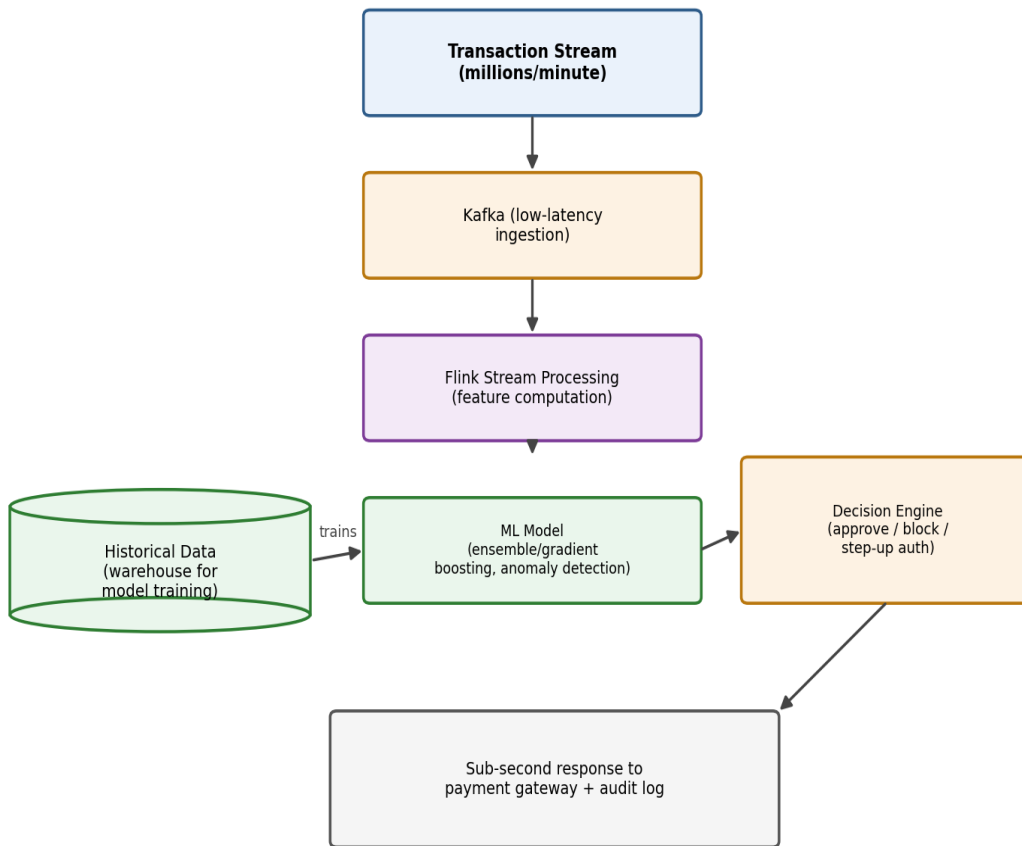
Understanding the industry context

Banking fraud detection is uniquely latency-sensitive: a decision must be made within the transaction's authorization window (typically under a second), and both false negatives (missed fraud) and false positives (blocking legitimate customers) carry significant financial and reputational cost, so the pipeline must be both extremely fast and highly accurate.

Big data pipeline design

- Ingestion: Kafka provides low-latency, high-throughput ingestion capable of handling millions of transaction events per minute.
- Stream processing: Apache Flink computes real-time features per transaction (velocity of transactions, deviation from typical spending pattern, geographic anomalies) with sub-second processing latency.
- Model training (offline): historical transaction data in a data warehouse trains ensemble models (gradient boosting such as XGBoost) or anomaly-detection models (autoencoders, isolation forests) on labeled fraud/non-fraud data.
- Real-time scoring: the trained model scores each incoming transaction's fraud risk using the features computed by the stream processor, without needing to retrain in real time.
- Decision engine: transactions are approved, blocked, or routed to step-up authentication (OTP/biometric) based on the risk score, with the response returned to the payment gateway within the authorization window.

Q4 — Banking Real-Time Fraud Detection Pipeline



Technologies and algorithms

Kafka and Flink provide the low-latency backbone; gradient boosting models are favored for their strong accuracy on structured tabular transaction features, while anomaly-detection techniques catch novel fraud patterns not well represented in historical labeled data. The separation of offline training from real-time scoring keeps the live path fast, since the computationally expensive part (training) happens outside the critical latency window.

Security and risk awareness

The fraud model itself is a sensitive asset — if compromised or reverse-engineered, fraudsters could learn to evade it — so model access and the feature-engineering logic must be tightly controlled. All decisions must be logged immutably for regulatory audit and dispute resolution, and the system must have a fallback/manual-review path for cases where the automated model is uncertain.

Q5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data from sensors and drones providing real-time field information, generated from multiple distributed sources, with farmers wanting insights to improve yield and resource usage.

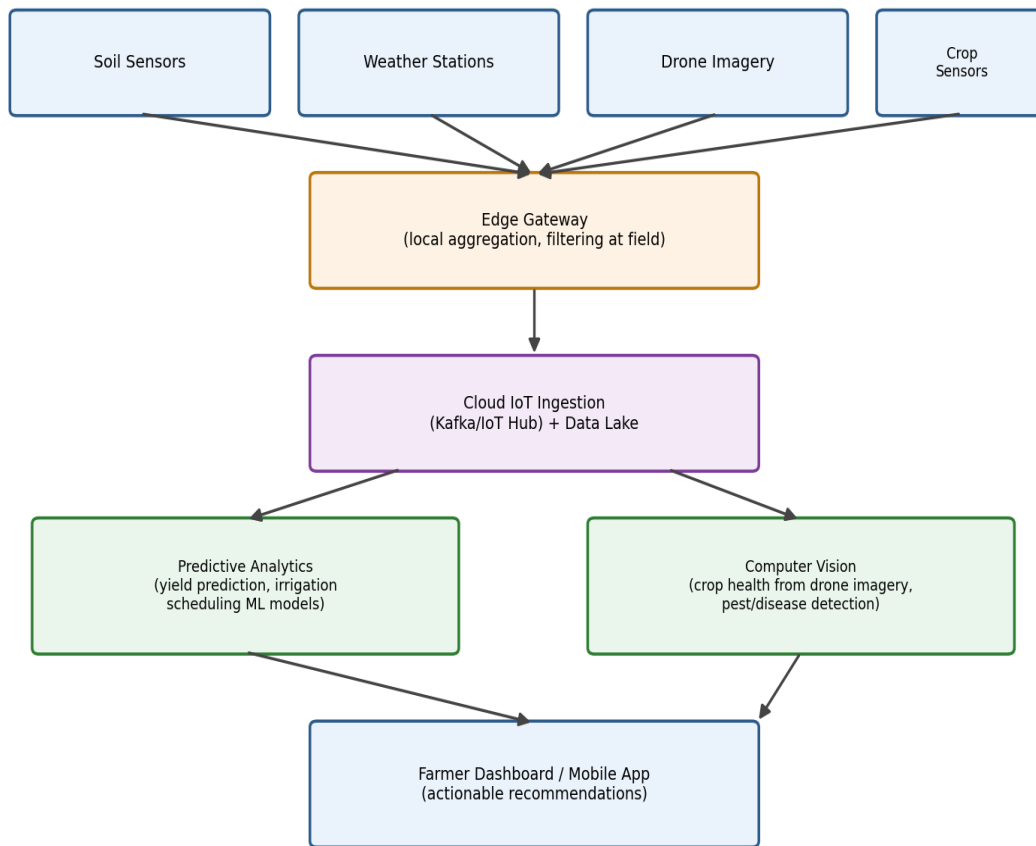
Understanding the industry context

Precision agriculture data originates from many geographically distributed, often connectivity-constrained field devices (soil sensors, weather stations, drones), so the solution must handle intermittent connectivity and bandwidth constraints while still delivering timely, actionable insights to farmers who may have limited technical background.

Big data solution design

- Edge processing: an edge gateway at the farm aggregates and filters raw sensor/drone data locally, reducing the volume that needs to be transmitted and allowing basic alerts to work even with intermittent connectivity.
- Cloud ingestion: filtered data is sent to the cloud via an IoT ingestion service (Kafka/IoT Hub) and stored in a data lake for both real-time and historical analysis.
- Predictive analytics: machine learning models trained on historical soil, weather, and yield data predict optimal irrigation schedules, fertilizer needs, and expected yield.
- Computer vision: drone imagery is processed with computer vision models to detect crop health issues, pest infestations, or disease symptoms across large fields faster than manual inspection.
- Farmer-facing delivery: a mobile app/dashboard translates model outputs into simple, actionable recommendations (e.g., "irrigate zone 3 tomorrow") rather than raw data or technical metrics.

Q5 — Precision Agriculture Big Data Solution



How analytics supports decision-making

Predictive analytics turns raw sensor readings into forward-looking guidance (e.g., predicting water stress before crops visibly suffer), while computer vision on drone imagery can catch localized problems (a pest outbreak in one section of a large field) that would be impractical to detect through manual walking inspection — together enabling farmers to intervene earlier and more precisely, improving yield while reducing wasted water and chemical inputs.

Security and risk awareness

Field sensor networks are often less physically secured than enterprise IT systems, making them a potential entry point for tampering (e.g., falsified sensor readings) or device compromise; the platform should authenticate each device, validate sensor data for plausibility before it drives automated actions (like irrigation triggers), and ensure farmers retain ownership and control over their own farm data rather than it being locked into a single vendor's platform.